

DM-seq Method

Note: *For reference only when preparing manuscripts. Please refer to the final analysis report for detailed content and methods.*

1. Experimental Procedure

1.1 Sample Quality Control

Please refer to QC report for methods of sample quality control

1.2 Library Construction, Quality Control and Sequencing

After sample qualification, 10 ng of genomic DNA was mixed with methylated lambda DNA. The mixture was then fragmented to 200–400 bp using the Covaris S220 ultrasonicator (Covaris, USA) (if the sample is cfDNA, fragmentation is not required; instead, fragmented methylated lambda DNA is mixed with the sample). Following fragmentation, adapter ligation was performed. Subsequently, the DNA fragments were treated using a proprietary enzymatic reaction to directly convert methylated cytosine into thymine. Finally, PCR amplification was carried out to complete the library construction. Library quality was assessed on the Agilent 5400 system (Agilent, USA) and quantified by QPCR, with a required concentration greater than 1.5 nM. Qualified libraries were subjected to paired-end sequencing on the Illumina platform (Illumina, USA).

2. Bioinformatics Analysis

2.1 Data Quality Control

First, evaluate the quality of raw data from the Illumina platform using FastQC (*fastqc* 0.11.8). Then filter the raw data with fastp (fastp 0.23.1) to obtain clean data. Reassess the quality of clean data using FastQC; these data are used for subsequent analysis.

2.2 Reference Genome Alignment

The reference genome and annotation files are downloaded directly from genome websites. Align clean reads to the reference genome using BWA (v 0.7.12) (Vasimuddin et al., 2019). Filter the reads to retain high-quality alignments ($\text{MAPQ} \geq 13$), and use uniquely mapped, PCR-duplicate-removed reads for further analysis.

2.3 Methylation Level Analysis

Methylation levels are analyzed using Rastair for animal samples and Astair for plant samples (Liu, Y et al., 2019).

2.4 Differential Methylation Analysis

Identify differentially methylated regions (DMRs) using the DSS software (version 2.12.0; Feng et al., 2014; Park et al., 2016; Wu et al., 2015). DSS uses a novel dispersion shrinkage method to estimate dispersion parameters via Gamma-Poisson or Beta-Binomial

distributions. Based on genomic distribution, DMRs are classified into gene coding regions (from TSS to TES) and promoter regions (2 kb upstream of TSS).

2.5 GO/KEGG Enrichment Analysis

Perform GO (Gene Ontology) enrichment analysis for DMR-associated genes using the R package Goseq (Young et al., 2010). Enrichment is considered significant when the p-value < 0.05 (corrected for gene length bias). Perform KEGG pathway enrichment analysis for DMR-associated genes using KOBAS (version 3.0; Mao et al., 2005).

2.6 Variant Detection Results

Single nucleotide polymorphism (SNP) refers to DNA sequence polymorphisms caused by single nucleotide variations at the genomic level, including base transitions and transversions. InDel (Insertion and Deletion) refers to small insertions and deletions. Insertions or deletions in coding regions or splice sites may alter protein translation. Frameshift variants occur when the length of the inserted/deleted base string is not a multiple of three, potentially shifting the entire reading frame. Frameshift variants have a greater impact on gene function and undergo stronger selective pressure in populations than non-frameshift variants. Detect individual SNPs and InDels using the TVC software with parameters: `--min-bq 20 || --min-mapq 1 || --min-depth 2 || --end-of-read-cutoff 5 || --error-rate 0.005 || --min-ao 2`.

2.7 SV Detection Results

Structural variation (SV) refers to large-scale insertions, deletions, inversions, translocations, and other sequence alterations at the genomic level. Detect insertions (INS), deletions (DEL), inversions (INV), intra-chromosomal translocations (ITX), and inter-chromosomal translocations (CTX) between samples and the reference genome using BreakDancer, based on pair-end read alignment relationships and actual insert size. Filter SV results supported by fewer than 2 PE reads.

2.8 CNV Detection Results

Copy number variation (CNV) refers to gains or losses of genomic fragment copy numbers. Detect CNVs using CNVnator (parameter: `-call 100`) by analyzing read depth across the genome to identify potential deletions and duplications.

2.9 Annotation

ANNOVAR is an efficient software tool for functional annotation of genetic variants detected from multiple genomes using up-to-date information. Given the chromosome, start/end positions, reference nucleotide, and variant nucleotide, ANNOVAR performs gene-based annotation, region-based annotation, filter-based annotation, and other functionalities.

References

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